

Heart Failure: Common Complications and Conditions

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Care
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[Link to Codes](#)

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Prevention

- Prevention issues and possible preventive measures include(1)(2)(3)(4)(5)(6)(7)(8):
 - ☐ Maintaining awareness of and treating patient characteristics and situations that pose high risk for development of heart failure; examples include:
 - Worsening renal function(9)
 - Ischemic heart disease, including prior myocardial infarction(10)
 - Anemia(11)
 - Use of cardiotoxic drugs (eg, anthracycline chemotherapy, immune checkpoint inhibitor), or medications causing fluid retention (eg, corticosteroids, NSAIDs, alpha-1 blockers)(12)
 - Need for aggressive fluid resuscitation (eg, sepsis, hemorrhage)(12)
 - Structural heart disease; examples include(13):
 - Valvular heart disease
 - Left ventricular hypertrophy
 - Hypertrophic cardiomyopathy
 - Infiltrative cardiomyopathy (eg, cardiac sarcoidosis, amyloidosis, hemochromatosis)
 - Dilated cardiomyopathy
 - Congenital heart disease(14)(15)
 - ☐ Providing meticulous perioperative care, with special attention to(16):
 - Fluid balance
 - Preoperative evaluation of left ventricular function
 - Recognition and treatment of underlying coronary artery disease
 - Optimization of heart failure therapy
 - Continuation of beta-blocker therapy[A](17)

Clinical Indications for Inpatient Care

- Ongoing inpatient care is indicated for **1 or more** of the following(1)(2)(3)(5)(6)(8):
 - Hemodynamic instability
 - Acute myocardial ischemia causing or consequence of heart failure(10)(18)(19)
 - New or worsening pulmonary edema
 - Hypoxemia
 - Tachypnea
 - Syncope
 - Altered mental status
 - Increased creatinine with reduction of more than 50% in estimated glomerular filtration rate from baseline
 -  [eGFR - Adult Calculator](#)
 - Progressively (ongoing) rising creatinine with reduction of more than 25% in estimated glomerular filtration rate from baseline
 -  [eGFR - Adult Calculator](#)
 - Massive skin edema or Anasarca with complications (eg, tissue breakdown with infection, inability to void due to edema)(3)

- o Pulmonary artery catheter monitoring needed[B]

Alternatives to Inpatient Care

- Alternatives to inpatient stay extension include(1)(2)(8)(23)(24)(25)(26):
 - o Outpatient care(27)(28)
 - Medications for heart failure (eg, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), diuretics, beta-blocker, aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide))[C][D](1)(26)(29)(30)(31)
 - Antihypertensive medication[E]
 - Salt and fluid restriction
 - Patient education and weight monitoring
 - Cardiac rehabilitation(32)(33)(34)
 - o Home care(35)(36)(37)
 - Medications for heart failure (eg, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), diuretics, beta-blocker, aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide))[C][D](1)(29)(30)(31)
 - Antihypertensive medication[E]
 - Salt and fluid restriction
 - Nursing visit for assessment
 - Laboratory testing
 - Multidisciplinary team care with intensive home care support(1)(38)(39)
 - Home-based cardiac rehabilitation(32)(40)
 - Patient education and weight monitoring
 - Palliative care(1)(29)(41)
 - o Skilled nursing facility(42)(43)
 - Medications for heart failure (eg, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), diuretics, beta-blocker, aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide))[C][D](1)(29)(30)(31)
 - Antihypertensive medication[E]
 - Nursing assessments
 - Laboratory testing and correction of electrolytes
 - Oxygen therapy
 - Monitoring weight and fluid balance
 - o Recovery facility(43)
 - Medications for heart failure (eg, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), diuretics, beta-blocker, aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide))[C][D](1)(29)(30)(31)
 - Antihypertensive medication[E]
 - Nursing assessments
 - Laboratory testing and correction of electrolytes
 - Parenteral medication
 - Oxygen therapy
 - Monitoring weight and fluid balance

Hospital Care

- Evaluation may include(1)(2)(3)(5)(8)(26)(44)(45)(46)(47):
 - o Laboratory testing, including CBC, electrolytes, fasting glucose, glycosylated hemoglobin, kidney function, urinalysis, liver function, lipid profile, transferrin, HIV, and thyroid function
 - o Cardiac biomarkers (eg, B-type natriuretic peptide, N-terminal pro-B-type natriuretic peptide level, and Troponin T or I)(48)(49)
 - o Vitamin D level, iron studies(50)(51)
 - o Chest x-ray
 - o ECG
 - o Echocardiogram
 - o Oxygen saturation or ABG
 - o Pulmonary artery catheter[B]
 - o Cardiology consultation(1)(52)(53)
 - o Stress echocardiogram or nuclear stress test imaging(18)
 - o Radionuclide ventriculography
 - o Ventricular viability testing (eg, nuclear study or cardiac MRI, for patients with ischemic cardiomyopathy)(18)
 - o Palliative care consultation(1)(29)(41)

- o Cardiac rehabilitation consultation(33)(54)
 - o Smoking cessation counseling(2)
 - o Nutrition counseling(1)
 - o Sleep evaluation(55)
- Treatment may include(1)(2)(3)(5)(8)(44)(45):
 - o Medications for heart failure (eg, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), diuretics, beta-blocker, aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide))[C][D](1)(29)(30)(31)
 - o Antihypertensive medication[E]
 - o Intravenous inotropic agent (eg, dobutamine, milrinone, for shock, end organ dysfunction, persistent symptoms, or for palliative care)(1)(37)
 - o Thromboembolic prophylaxis
 - o Intravenous iron(11)(51)(56)
 - o Thoracentesis if significant pleural effusion present
 - o Noninvasive ventilation or continuous positive airway pressure(57)
 - o Discontinuation of cardiotoxic medications, if possible(12)
 - o Adjustment of drug dosages for primary diagnosis as needed for level of cardiac function
 - o Treatment of myocardial ischemia. See Angina [ISC](#) or Myocardial Infarction [ISC](#) guideline as clinically appropriate.
 - o Treatment of hypertension with heart-failure-appropriate antihypertensive medications(58)(59)
 - o Treatment of arrhythmia(52)(60)
 - o Mechanical circulatory support (eg, ventricular assist device)(21)(22)(61)(62)
 - o Ultrafiltration for diuresis resistant fluid overload(63)
 - o Treatment of metabolic abnormalities. See Electrolyte Disorder: Common Complications and Conditions [ISC](#) guideline as appropriate.
 - Intravenous or oral repletion of electrolytes
 - For acidosis: citric acid and sodium citrate (Bicitra), bicarbonate, dialysis
 - For alkalosis: potassium repletion, carbonic anhydrase inhibitor

Discharge

- Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present(1)(2)(5)(64)(65):
 - o Hemodynamic stability
 - o Tachypnea absent
 - o Oxygenation at baseline or acceptable for next level of care
 - o Dyspnea at baseline or acceptable for next level of care
 - o Cardiac rate and rhythm acceptable
 - o No evidence of myocardial ischemia
 - o Pulmonary edema absent or acceptable for next level of care
 - o Peripheral or sacral edema absent or acceptable for next level of care
 - o Mental status at baseline
 - o Volume status acceptable on oral medication
 - o Renal function at baseline or acceptable for next level of care
 - o Electrolyte abnormalities absent or acceptable for next level of care
 - o Precipitating factors absent or controlled
 - o Medication regimen established for next level of care[C][D]
 - o Patient or caregiver has received written discharge instructions or educational material addressing activity level, diet, discharge medications, follow-up appointment, weight and edema monitoring, dietary and fluid restriction instructions, and what to do if symptoms worsen.(2)(65)(66)

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Footnotes

[A] Long-term beta-blocker therapy may often be continued in hospitalized patients with acute exacerbations of COPD.(17) [A in Context Link 1]

[B] Pulmonary artery catheterization (right-sided heart catheterization) may be needed when diagnosis is unclear (eg, suspected heart failure with left ventricular ejection fraction greater than 40%), for patients with ongoing symptoms or uncertain fluid status following initial diuretic treatment, or to guide device selection and utilization of mechanical circulatory support.(1)(20)(21)(22) [B in Context Link 1, 2]

[C] A specialty clinical guideline based upon a systematic review of the evidence concludes that heart failure patients with evidence of significant fluid overload should be promptly treated with intravenous loop diuretics (eg, initial dose twice the home regimen in terms of oral to parenteral milligram equivalents) and titrated to resolution of clinical signs and symptoms of congestion (eg, serial doubling of dosage as bolus or continuous infusion).(1) This guideline further recommends increasing the diuretic dose or adding a second diuretic (eg, thiazide diuretic or aldosterone antagonist with diuretic properties) until an effective regimen is identified that is adequate to relieve congestion.(1) [C in Context Link 1, 2, 3, 4, 5, 6]

[D] A specialty guideline recommends 4 medication classes as guideline-directed medical therapy (GDMT) for patients with heart failure with reduced ejection fraction (HFrEF): beta-blockers, renin-angiotensin inhibitors (angiotensin receptor-neprilysin inhibitor preferred), aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors.(1) This same guideline recommends the following for patients hospitalized with HFrEF absent clear contraindications: pre-existing GDMT should be continued and optimized (titration to therapeutic dose begun as inpatient); GDMT should not be discontinued due to mild decrease in renal function or blood pressure; GDMT should be initiated while hospitalized when clinical stability achieved (if all drug classes not on board); and if GDMT is discontinued, it should be reinstated as soon as possible.(1) Vericiguat further reduces the risk of adverse cardiac outcomes in heart failure and may either be started with the standard quadruple therapy outlined above for severe worsening heart failure, used as a substitute agent for patients with contraindication or intolerance to one or more standard medications, or used as an add-on therapy for inadequate response to optimal quadruple therapy.(26) An expert specialty society panel also recommends these 4 medication classes as GDMT for heart failure with preserved ejection fraction (HFpEF).(29) [D in Context Link 1, 2, 3, 4, 5, 6]

[E] Additional hypertensive agents beyond the recommended heart failure guideline-directed medical therapy (GDMT) medications may be needed for patients with heart failure with preserved ejection fraction (HFpEF) to attain a goal systolic blood pressure less than 130 mm Hg.(29) [E in Context Link 1, 2, 3, 4, 5]

Definitions

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

References

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Anasarca

- Anasarca is generalized edema that is widespread and may be more pronounced in dependent areas (eg, back while lying down, legs while sitting up).(1)

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Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
 - Vital sign abnormality not readily corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
 - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
 - Hypotension that persists despite appropriate treatment (eg, volume repletion, treatment of underlying cause)
 - Orthostatic hypotension that persists despite appropriate treatment (eg, volume repletion)
 - Hypotension that is severe, as indicated by **ALL** of the following:
 - Hypotension
 - Severe hypotension, as indicated by **1 or more** of the following:
 - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
 - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
 - Mean arterial pressure[A][D] less than 65 mm Hg
 - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

References

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2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-114.
3. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.
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5. Nyhan A. Cardiology. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:145-188.e4.
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- American Heart Association (AHA), European Society of Cardiology (ESC) Association for Acute Cardiovascular Care (ACVC), International Society for Heart and Lung Transplantation (ISHLT), Society of Critical Care Medicine (SCCM), and Society of Thoracic Surgeons (STS) in December 2021. *Journal of the American College of Cardiology* 2022;Online. DOI: 10.1016/j.jacc.2022.01.018.
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 10. Kraut JA, Madias NE. Lactic acidosis. *New England Journal of Medicine* 2014;371(24):2309-2319. DOI: 10.1056/NEJMr1309483.
 11. Olivieri L, Chasm R. Diabetic ketoacidosis in the pediatric emergency department. *Emergency Medicine Clinics of North America* 2013;31(3):755-773. DOI: 10.1016/j.emc.2013.05.004.
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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)
- C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated. (12)
- D. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals 1/3 SBP + 2/3 DBP.(1)(4) A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

1. Puskasich MA, Jones AE. Shock. In: Walls RM, editor. *Rosen's Emergency Medicine: Concepts and Clinical Practice*. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:34-41.e1.
2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. *Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics*. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-114.
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5. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). *Journal of the American Medical Association* 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.

Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older^[A]
 - Mean arterial pressure^{[A][B]} less than 70 mm Hg in adult or child 10 years or older
 - Decrease in baseline mean arterial pressure of 25% or more,^{[A][B]} with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
 - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age^[A]
 - SBP less than 70 mm Hg in infant 1 to 11 months of age^[A]

References

- Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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- Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

- Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent^[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

- Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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- Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

- Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following⁽¹⁾:
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air^{[A][B]}
 - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)^[A]
 - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

References

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Footnotes

- A. Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.⁽²⁾ Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.⁽²⁾ Specific target values may require adjustment when care is delivered at high altitude.⁽¹⁾⁽³⁾
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Orthostatic hypotension

- Orthostatic hypotension,^{[A][B]} as indicated by **1 or more** of the following⁽¹⁾⁽²⁾⁽³⁾:
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

- Shibao C, Lipsitz LA, Biaggioni I, American Society of Hypertension Writing Group. Evaluation and treatment of orthostatic hypotension. Journal of the American Society of Hypertension 2013 Jul-Aug;7(4):317-324. DOI: 10.1016/j.jash.2013.04.006.
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Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.⁽¹⁾⁽²⁾⁽³⁾
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Tachycardia

- Tachycardia^{[A][B]} as indicated by **1 or more** of the following⁽¹⁾⁽²⁾:
 - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
 - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
 - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
 - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
 - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
 - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia^{[A][B]} absent, as indicated by **1 or more** of the following⁽¹⁾⁽²⁾:
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the

term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea,[A][B] as indicated by respiratory rate of **1 or more** of the following(2)(3):
 - Greater than 18 breaths per minute in adult or child 13 years of age or older[A]
 - Greater than 22 breaths per minute in child 6 to 12 years of age[A]
 - Greater than 25 breaths per minute in child 3 to 5 years of age[A]
 - Greater than 30 breaths per minute in child 1 or 2 years of age[A]
 - Greater than 40 breaths per minute in infant 6 to 11 months of age[A]
 - Greater than 45 breaths per minute in infant 3 to 5 months of age[A]
 - Greater than 60 breaths per minute in infant 1 or 2 months of age[A]

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
3. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(1) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(1)

Tachypnea absent

- Tachypnea[A][B] absent, as indicated by respiratory rate of **1 or more** of the following(2)(3):
 - Less than or equal to 18 breaths per minute in adult or child 13 years of age or older
 - Less than or equal to 22 breaths per minute in child 6 to 12 years of age
 - Less than or equal to 25 breaths per minute in child 3 to 5 years of age
 - Less than or equal to 30 breaths per minute in child 1 or 2 years of age
 - Less than or equal to 40 breaths per minute in infant 6 to 11 months of age
 - Less than or equal to 45 breaths per minute in infant 3 to 5 months of age
 - Less than or equal to 60 breaths per minute in infant 1 or 2 months of age

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.⁽¹⁾ This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.⁽¹⁾

Troponin

- Troponin elevation alone does not indicate acute myocardial infarction; factors to consider include⁽¹⁾⁽²⁾:
 - Patients can have elevated levels of cardiac troponin due to other causes of myocardial injury, either acutely or chronically (eg, heart failure, myocarditis, pulmonary embolism, cardiomyopathy, defibrillator shock).
 - Patients with an elevated troponin should, on subsequent measurement, have a fall or rise in the troponin level of more than 20% to confirm acuity of myocardial injury. Absent this variation, the acuity of the myocardial injury should be questioned.
 - Attributing acute troponin release to myocardial ischemia depends on the presence of associated symptoms (eg, chest pain, dyspnea), ECG changes (eg, acute ST-segment deviation), and imaging findings (eg, new wall motion abnormalities on echocardiogram).
 - A report from a facility (eg, hospital) laboratory of an elevated troponin level, in general, indicates a level above the 99th percentile for the specific troponin assay used. The precise value that represents the 99th percentile depends on the specific troponin assay used.
 - The recommended time interval between serial troponin level assessments (eg, to detect fall or rise in troponin level) is 1 to 3 hours for high-sensitivity troponin, and 3 to 6 hours for conventional (contemporary) troponin assays.
 - Evidence-based specialty society guidelines recommend use of high-sensitivity troponin assays, rather than older less-sensitive assays (collectively labeled contemporary assays).⁽¹⁾⁽³⁾ This is due to more rapid detection or exclusion of myocardial injury and increased diagnostic accuracy with high-sensitivity assays.⁽¹⁾⁽³⁾

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Codes

ICD-10 Diagnosis: I09.81, I11.0, I13.0, I13.2, I50.1, I50.20, I50.21, I50.22, I50.23, I50.30, I50.31, I50.32, I50.33, I50.40, I50.41, I50.42, I50.43, I50.814, I50.82, I50.83, I50.84, I50.89, I50.9, T86.22 [Hide]

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